Akoizo Original Practice - Form 2 - Answer key

Division B | Dynamic Planet | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Snow depth equals liquid-water depth automatically?

Unit: Freshwater distribution, precipitation, and climate.
Worked-case data: Two storms deliver equal rainfall but one lasts twice as long in the lab. Compare volume and peak.

A. Only in mountains

B. Only in spring

C. No

D. Always

Correct: C

2. (2 points) Dissolved load consists of…

Unit: Sediment transport and channel forms.
Worked-case data: A river has Q=20 m^3/s. Compare velocity for areas 10 and 20 m^2.

A. Only gravel

B. Only floating wood

C. Only sand

D. Ions and molecules in solution

Correct: D

3. (2 points) Hydrograph area represents…

Unit: Streamflow, hydrographs, and floods.
Worked-case data: A triangular hydrograph has base 4 h and peak 10 m^3/s above zero. Find volume.

A. Volume

B. Only peak stage

C. Only velocity

D. Only basin area

Correct: A

4. (2 points) Every lake mixes twice per year?

Unit: Lakes, wetlands, and freshwater storage.
Worked-case data: A storage feature lowers peak outflow but releases the same event volume later. Was water destroyed?

A. Only if freshwater

B. No

C. Always

D. Only if deep

Correct: B

5. (2 points) A contour connects equal…

Unit: Maps, monitoring, and integrated freshwater investigations.
Worked-case data: Two wells have heads 110 and 106 m, separated by 200 m. Find gradient magnitude.

A. Rainfall always

B. Chemical concentration always

C. Elevation

D. Discharge

Correct: C

6. (2 points) A one-event lab predicts climate?

Unit: Freshwater distribution, precipitation, and climate.
Worked-case data: Two storms deliver equal rainfall but one lasts twice as long in the lab. Compare volume and peak.

A. No

B. Always

C. Only with more sliders

D. Only for wet regions

Correct: A

7. (2 points) Competence describes…

Unit: Sediment transport and channel forms.
Worked-case data: A river has Q=20 m^3/s. Compare velocity for areas 10 and 20 m^2.

A. Only rainfall

B. Largest transportable particles

C. Total water volume

D. Only sediment color

Correct: B

8. (2 points) A 100-year label means exact spacing?

A. Only for rivers

B. Only after one flood

C. No

D. Always

Correct: C

9. (2 points) A bog is typically associated with…

Unit: Lakes, wetlands, and freshwater storage.
Worked-case data: A storage feature lowers peak outflow but releases the same event volume later. Was water destroyed?

A. Only seawater tides

B. Only bedrock magma

C. Only industrial pipes

D. Precipitation-dominated water supply

Correct: D

10. (2 points) Vertical exaggeration changes…

Unit: Maps, monitoring, and integrated freshwater investigations.
Worked-case data: Two wells have heads 110 and 106 m, separated by 200 m. Find gradient magnitude.

A. Displayed slope appearance

B. Actual geology automatically

C. Measured head automatically

D. Water mass

Correct: A

11. (2 points) Equal annual rain implies equal reliable supply?

Unit: Freshwater distribution, precipitation, and climate.
Worked-case data: Two storms deliver equal rainfall but one lasts twice as long in the lab. Compare volume and peak.

A. Only near oceans

B. Only for groundwater

C. No

D. Always

Correct: C

12. (2 points) Equal discharge guarantees equal velocity?

Unit: Sediment transport and channel forms.
Worked-case data: A river has Q=20 m^3/s. Compare velocity for areas 10 and 20 m^2.

A. Always

B. Only for rivers

C. Only at flood stage

D. No

Correct: D

13. (2 points) Stage and discharge are…

Unit: Streamflow, hydrographs, and floods.
Worked-case data: A triangular hydrograph has base 4 h and peak 10 m^3/s above zero. Find volume.

A. Related but different quantities

B. Identical units

C. Always equal numbers

D. Unrelated by any model

Correct: A

14. (2 points) Wetlands remove all contaminants forever?

Unit: Lakes, wetlands, and freshwater storage.
Worked-case data: A storage feature lowers peak outflow but releases the same event volume later. Was water destroyed?

A. Only if large

B. No

C. Always

D. Only in summer

Correct: B

15. (2 points) Best monitoring design begins with…

Unit: Maps, monitoring, and integrated freshwater investigations.
Worked-case data: Two wells have heads 110 and 106 m, separated by 200 m. Find gradient magnitude.

A. Only convenience

B. Only an attractive map

C. A defined question

D. Maximum instruments

Correct: C

16. (4 points) Why can runoff lag snowfall?

Unit: Freshwater distribution, precipitation, and climate.
Worked-case data: Two storms deliver equal rainfall but one lasts twice as long in the lab. Compare volume and peak.

answer: Snow stores water until melt. (4 points).

17. (4 points) Why can a sediment-starved flow erode?

Unit: Sediment transport and channel forms.
Worked-case data: A river has Q=20 m^3/s. Compare velocity for areas 10 and 20 m^2.

answer: Transport ability can exceed supplied load, allowing bed or bank entrainment. (4 points).

18. (4 points) Why can a rating curve change?

Unit: Streamflow, hydrographs, and floods.
Worked-case data: A triangular hydrograph has base 4 h and peak 10 m^3/s above zero. Find volume.

answer: Channel geometry or backwater conditions can alter stage-flow relationships. (4 points).